Richard Harnisch

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Education

MSc Artificial Intelligence, current GPA 9.0 September 2026–2028
University of Amsterdam

- Focusing on Reinforcement Learning and LLMs, working on extracurricular project with **Pangram** on **LLM detection**.

BSc Artificial Intelligence, cum laude, GPA 8.4/10 2022–2026
University of Groningen

- Thesis: Interpolation Threshold and **Double Descent in RL**. Tracked Fisher Information Matrix trace across over-parameterized agents; found local curvature can decouple from the generalization gap under policy-induced non-stationarity.
- Formal math/CS foundations: Calculus for Artificial Intelligence; Linear Algebra and Multivariable Calculus; Statistics; Analysis; Probability Theory; Probability and Measure; Linear Algebra 2; Partial Differential Equations; Imperative Programming; Algorithms and Data Structures; Object-Oriented Programming.
- Advanced AI/ML: Applied Machine Learning, **Reinforcement Learning**, Uncertainty in ML, Machine Translation (**9.5 each**); Neural Networks, Natural Language Processing (9 each); Advanced Logic (8.5).

International Baccalaureate; German Abitur 1.0 2020–2022
Dresden International School 42/45 points

Higher Level Mathematics, Physics, Economics; **maximum marks** in Mathematics, Physics, German, Spanish. Extended Essay on public-key cryptography including Euler's totient function, Chinese remainder theorem & vulnerabilities.

Research and ML Experience

Research Assistant: Data Center Optimization 2026–present
Multiscale Networked Systems Group Amsterdam

- Working at the University of Amsterdam under Dr. Shashikant Ilager on Data Center design, targeting publication.
- Optimizing electricity use and carbon intensity while keeping performance within threshold values for both inference and training workloads. Exploring **agent-driven design** for physically verified data-center designs while minimizing cost.

Data Science Intern: Electricity Price Forecasting 2025–2026
Nederlandse Gasunie Groningen/Remote

- Developed a day-ahead electricity price forecasting system using SARIMA, LSTMs, and Transformers in 5-person team.
- Designed and deployed **distribution-shift monitoring** over ENTSO-E price data and Dutch grid-production data; system alerts operators and triggers retraining under covariate drift; worked **directly with Chief Data Scientist**.
- Conducted feature engineering and data management for **real-world forecasting** in a grid undergoing renewable- and geopolitics-driven distribution shifts including alerting operator upon distribution shift to retrain model on new data.

Bachelor's Thesis: Double Descent in Reinforcement Learning 2025–2026
University of Groningen Supervisor: Dr. Matthia Sabatelli

- Investigated the **interpolation threshold** and Double Descent phenomenon in **over-parameterized Reinforcement Learning agents** under non-stationary data distributions created by **randomized environments** and adaptive policy.
- Tracked the Fisher Information Matrix (FIM) trace as a **generalization metric** across constrained policy optimizers (TRPO) at varying model capacity and training durations. Gained experience with **high-performance computing**.
- Found empirical evidence that FIM decorrelates from the generalization gap under policy-induced non-stationarity: constrained optimizers exhibit expected FIM spike near the interpolation threshold, but the curvature signal does not predict generalization gap evolution, challenging the applicability of classical information-geometric bounds in RL.
- Raises open questions about the role of implicit regularization and optimizer geometry in non-stationary generalization.

Machine Learning Engineer and Software Developer 2024–2026
descript GmbH Dresden/Remote

- Independently **led industrial ML research collaboration with TU Dresden** on textile defect detection; defined project methodology for learning from limited data and built **active learning** system for prioritized expert review.
- Led **full SWE pipeline for production features** from issue to deployment, such as a building management scheduler.

Teaching Assistant: Introduction to Machine Learning 2025–2026
University of Groningen

- Led lab sessions and graded assignments for 100 second-year AI students, communicated ML concepts to broad audience.
- Topics: ML paradigms, model selection, regularization, evaluation metrics, data ingestion, processing and augmentation.

Selected Projects

Hackathon Winner. With two teammates, won AI Hackathon Haarlem 2026 with **agentic orchestration system** to ingest house style based on example `pptx` files and delivers arbitrary editable `pptx` decks including speaker notes & handouts.
Study Association Board. As board of **1,000+ member AI/CS study association** at University of Groningen **organized AI symposium** with academic & industry speakers; ~150 attendees. Organized international excursions.
Mars Rover Competition. Authored technical report for European Space Agency rover competition and presented with MakerCie team; placed **1st among 33 teams** and received maximum points for presentation.
Open Source Contributions. Submitted bug-fix PRs to both **Opencode** and **PyTorch (vision)**, accepted and merged.
Languages: English (TOEFL iBT 6/6, C2) and **German** native; **Dutch** and **Spanish** conversational.